Credit Risk ML Capstone

Viva Preparation Guide · 30 Questions

Important project-scope note

The uploaded presentation is titled “Credit Card Fraud Detection,” but its workflow, target definition, feature names (for example, DAYS_BIRTH and AMT_CREDIT), and results describe loan default / credit-risk prediction. These questions therefore follow the evidence in the slides rather than silently treating the data as transaction-level card fraud. The model metrics below are reported in the presentation and should be verified against the actual code, data split, class labels, and leakage checks before being presented as independently validated results.

How to use this guide

Practice answering each question aloud in 30–60 seconds. Start with the direct answer, explain the technical reason, then connect it to business risk. For advanced questions, acknowledge limitations instead of overstating what the slides prove.

01 · BASIC

Foundations and project understanding

Q01. What is the objective of your project?

Model answer: The objective is to predict whether a loan applicant may become a defaulter (TARGET = 1) or a non-defaulter (TARGET = 0), using applicant information to support credit-risk assessment.

Q02. What type of machine-learning problem is this?

Model answer: It is a supervised binary-classification problem because the target has two labeled outcomes: 0 for non-defaulter and 1 for defaulter.

Q03. Which dataset does your presentation describe?

Model answer: The presentation describes a loan-default dataset with applicant demographic, employment, financial, and behavioral variables. It is a credit-default project, not a transaction-level credit-card-fraud dataset.

Q04. What does the TARGET column represent?

Model answer: TARGET = 0 represents a non-defaulter and TARGET = 1 represents a defaulter, according to the presentation.

Q05. Why is exploratory data analysis (EDA) important?

Model answer: EDA helps understand target prevalence, distributions, outliers, missingness, relationships between variables, and potential data-quality issues before modeling.

Q06. What is class imbalance, and why does it matter?

Model answer: Class imbalance occurs when target classes have very different frequencies. Accuracy can look high even if the model performs poorly on the minority class, so class-specific metrics are important.

Q07. What is Logistic Regression?

Model answer: Logistic Regression is a classification algorithm that estimates a probability for the positive class using a logistic function. A threshold converts the probability into a class prediction.

Q08. Why did you use Logistic Regression as a baseline?

Model answer: It is computationally efficient and relatively interpretable, and it provides a benchmark for judging whether more complex models deliver meaningful gains.

Q09. What is a confusion matrix?

Model answer: It summarizes predictions against actual labels using true positives, true negatives, false positives, and false negatives. For this project, the positive class is defaulter (1).

Q10. What is the difference between precision and recall?

> **Model answer:** Precision is the fraction of predicted defaulters who are actually defaulters. Recall is the fraction of actual defaulters correctly identified by the model.

02 · INTERMEDIATE

Workflow, preprocessing and model evaluation

Q11. Explain the workflow used in your project.

Model answer: The presentation shows data collection and target definition, EDA, preprocessing (outlier handling, categorical encoding, numerical scaling and target balancing), train/test splitting, model training, evaluation, hyperparameter tuning, a suitability decision, and deployment as a proposed next step.

Q12. Why should missing values be investigated before imputation?

Model answer: Missingness can reflect data collection issues or meaningful patterns. Inspecting its frequency and relationship to the target helps choose a defensible imputation strategy and avoid hiding data-quality problems.

Q13. How would you distinguish duplicate records from repeated legitimate observations?

Model answer: Check whether rows are exact duplicates and inspect transaction/application identifiers and domain rules. Repeated values alone do not prove duplication; remove records only when they are confirmed erroneous duplicates.

Q14. Why is data leakage dangerous?

Model answer: Leakage gives the model information unavailable at prediction time, making evaluation overly optimistic. Exclude post-outcome fields, chargeback/outcome information, and identifiers that encode the label; fit preprocessing only on training data.

Q15. Why should preprocessing be fitted only on the training set?

Model answer: Imputers, scalers, encoders, and resampling methods can learn information from data. Fit them within a pipeline on training folds, then apply the learned transformations to validation/test data without refitting.

Q16. Why might scaling be useful for Logistic Regression but less important for tree models?

Model answer: Scaling puts numerical features on comparable ranges and can help optimization and regularization in Logistic Regression. Tree splits depend mainly on ordering and are generally insensitive to monotonic feature scaling.

Q17. How do you interpret the Random Forest feature-importance chart?

Model answer: The presentation ranks DAYS_BIRTH, DAYS_EMPLOYED, and DAYS_LAST_PHONE_CHANGE highly, followed by variables such as AMT_GOODS_PRICE, AMT_CREDIT, and AMT_ANNUITY. Importance indicates model reliance, not causation or proof that a feature is fair or appropriate.

Q18. What does the correlation heatmap tell you, and what can it not tell you?

Model answer: It reveals pairwise linear associations among selected numerical variables. It does not establish causality and may miss nonlinear relationships; weak individual correlations with TARGET do not rule out useful combined predictive signal.

Q19. How should you interpret the reported model comparison?

Model answer: The slides report ROC-AUC values of 0.689 for Logistic Regression, 0.931 for baseline Random Forest, 0.951 for tuned Random Forest, and 0.970 for XGBoost. These figures suggest stronger ranking performance for XGBoost in the reported comparison, subject to checking that the evaluation split and procedure were consistent and leakage-free.

Q20. What is hyperparameter tuning, and why do it?

Model answer: It searches for model settings—such as tree depth, number of trees, or learning rate—that improve validation performance. Choose settings using training/validation data or cross-validation, not by repeatedly optimizing against the final test set.

03 · ADVANCED

Critical evaluation and production thinking

Q21. Why can a model with high accuracy still be unsuitable for default-risk screening?

Model answer: Accuracy can hide minority-class errors and does not encode the different costs of missed defaults and wrongly flagged good applicants. Evaluate default-class precision/recall, confusion matrix, ROC-AUC and preferably precision-recall performance, then connect thresholds to operational costs.

Q22. Explain ROC-AUC and how it differs from accuracy.

Model answer: The ROC curve plots true-positive rate against false-positive rate across thresholds. AUC summarizes ranking discrimination across thresholds; accuracy evaluates classifications at one selected threshold. A high AUC does not guarantee calibrated probabilities or an appropriate operating threshold.

Q23. How would you select a classification threshold for this project?

Model answer: Choose it on a validation set based on the relative costs of false negatives and false positives, available review capacity, and required recall/precision. Report the chosen threshold and its confusion matrix; do not select it using the final test set.

Q24. What does the threshold-0.6 Logistic Regression result tell you?

Model answer: The slide reports default-class precision 0.69 and recall 0.41, with 23,095 true defaulters detected and 33,442 missed in the displayed confusion matrix. It illustrates a trade-off at that threshold; the operating point should be assessed against business costs and checked against the exact evaluation setup.

Q25. Compare the reported baseline and tuned Random Forest results.

Model answer: The presentation reports accuracy of 0.82 for baseline Random Forest and 0.86 for tuned Random Forest. Default-class recall rises from 0.66 to 0.74, while precision is reported as 0.96 and 0.97 respectively. Confirm the class labels, threshold, and evaluation data are identical before making a direct comparison.

Q26. What is the main interpretation of the reported XGBoost result?

Model answer: The slide reports accuracy around 0.95, ROC-AUC 0.970, and default-class recall 0.90. The displayed confusion matrix contains 50,606 true defaulters detected and 5,931 missed, with 3 non-defaulters incorrectly flagged. These unusually strong results should prompt careful leakage and split checks before claiming production readiness.

Q27. Why is a random train/test split not always suitable for financial prediction?

Model answer: If observations have a time order, random splitting can let future patterns inform evaluation of past cases. A time-based or out-of-time test may better represent deployment. The appropriate split depends on how the data was collected and how predictions will be used.

Q28. How would you assess fairness when gender and regional variables appear predictive?

Model answer: Measure error rates and calibration across relevant groups, check data quality and legal/policy constraints, and discuss whether each feature is appropriate for the intended decision. Predictive association alone does not justify using a sensitive attribute; fairness evaluation requires context and governance.

Q29. How would you monitor the model after deployment?

Model answer: Track input-data quality, missingness, feature and score drift, default prevalence, calibration, class-specific performance once labels mature, subgroup errors, and operational outcomes. Define alert thresholds, ownership, review cadence, and a retraining/rollback process.

Q30. What limitations and next steps would you acknowledge in the viva?

Model answer: The slides do not document every implementation detail, validation protocol, calibration result, confidence interval, or deployment evidence. I would verify the reported metrics on a leakage-free holdout, document the split and threshold, compare precision-recall and cost-based outcomes, assess fairness and calibration, and treat deployment as future work until operational validation is complete.

REFERENCE

Quick metric reference

Metric	Meaning
Precision	Of predicted defaulters, the proportion who truly defaulted.
Recall / sensitivity	Of actual defaulters, the proportion correctly identified.
F1-score	Harmonic mean of precision and recall.
False-positive rate	Of actual non-defaulters, the proportion incorrectly flagged as defaulters.
ROC-AUC	Threshold-independent measure of how well scores rank positive cases above negative cases.